Priyan Vaithilingam

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Experience

Research Scientist, Apple — *Seattle, USA*
Apr 2025 - Present
- Generative UI – on the fly synthesis of UI, adapting to user intent and context.
- Building computer use agents that perceive, reason about, and act on graphical user interfaces.

Research Intern, Microsoft Research — *Redmond, USA*
May 2024 - Aug 2024
- Embedding AI assistance directly into OS-level interactions like Copy-Paste.
- Created MagicCopy – to helps users smartly copy-paste data between applications using LLMs.

Research Intern, Microsoft Research — *Redmond, USA*
May 2023 - Aug 2023
- Re-imagining interaction techniques for AI-assisted data visualization tools
- Focusing on dynamic UI for editing and interacting with visualizations that adapts to user intent.

Research Intern, Microsoft — *Remote*
Sept 2021 - Aug 2022
- Re-imagining interaction techniques for AI-based automated programming tools
- Research on improving Intellicode user experience in Visual Studio

Research Fellow, Microsoft Research India — *Bangalore, India*
Mar. 2016 - July 2018
- Honorable mention paper in CHI 2018 | Worldwide 2nd place in Microsoft Hackathon 2017.
- Research in Human Computer Interaction, Programming Languages and Information Retrieval.

Education

Harvard University — *Cambridge, USA*
PhD in Computer Science | CGPA: 4/4 | Advised by Dr. Elena L. Glassman — *July 2020 - March 2025*

University of California, San Diego — *San Diego, USA*
M.S. in Computer Science | GPA: 3.96/4 | Advised by Dr. Philip Guo — *Sept 2018 - June 2020*

Indian Institute of Technology (IIT), Indore — *Indore, India*
B.Tech. in Computer Science & Engineering | GPA: 9/10 | Advised by Dr. Gourinath Banda — *June 2011 - May 2015*

Publications

The Way We Notice, That's What Really Matters: Instantiating UI Components with Distinguishing Variations
CHI 2026
Priyan Vaithilingam, Alan Leung, Jeffrey Nichols, Titus Barik

Semantic Commit: Helping Users Update Intent Specifications for AI Memory at Scale
UIST 2025
Priyan Vaithilingam, Munyeong Kim, Frida-Cecilia Acosta-Parenteau, Daniel Lee, Amine Mhedhbi, Elena L. Glassman, Ian Arawjo

Imagining a Future of Designing with AI: Dynamic Grounding, Constructive Negotiation, and Sustainable Motivation
DIS 2024
Priyan Vaithilingam, Ian Arawjo, Elena L. Glassman

DynaVis: Dynamically Synthesized UI Widgets for Visualization Editing
CHI 2024 (**Best Paper Award**)
Priyan Vaithilingam, Elena L. Glassman, Jeevana Priya Inala, Chenglong Wang

ChainForge: A Visual Toolkit for Prompt Engineering and LLM Hypothesis Testing
CHI 2024 (**Honorable Mention Award**)
Ian Arawjo, Chelse Swoopes, **Priyan Vaithilingam**, Martin Wattenberg, Elena L. Glassman

Towards More Effective AI-Assisted Programming: A Systematic Design Exploration to Improve Visual Studio IntelliCode's User Experience
ICSE SEIP 2023
Priyan Vaithilingam, et al.

Expectation vs. Experience: Evaluating the Usability of Code Generation Tools Powered by Large Language Models
CHI LBW 2022
Priyan Vaithilingam, Tianyi Zhang, Elena L. Glassman

Assuage: Assembly Synthesis Using A Guided Exploration
UIST 2021
Jingmei Hu, **Priyan Vaithilingam**, Stephen Chong, Margo Seltzer, Elena L. Glassman

Interpretable Program Synthesis
CHI 2021
Tianyi Zhang, Zhiyang Chen, Yuanli Zhu, **Priyan Vaithilingam**, Xinyu Wang, Elena L. Glassman

Bespoke: Interactively Synthesizing Custom GUIs from Command-Line Applications By Demonstration
UIST 2019
Priyan Vaithilingam, Philip Guo

CodeTalk: Improving programming environment accessibility for Visually Impaired developers
CHI 2018 (**Honorable Mention Award**)
Venkatesh Potluri, **Priyan Vaithilingam**, Suresh Iyengar, Vidya Y, Manohar Swaminathan, Gopal Srinivasa

Patents

Content Management Tool for Content Transformation
US Patent 12,705,251 (2026)
J.P. Inala, G.A. Ramos, N.M. Riche, C. Wang, **P. Vaithilingam**

Software Development Artifact Name Generation
US Patent 12,487,817 (2025)
P. Groenewegen, **P. Vaithilingam**, R.J. Malpani, W. Wang

Dynamically Synthesized User Interface Widgets
US Patent App. 18/533,674 (2025)
J.P. Inala, C. Wang, **P. Vaithilingam**

Awards

Best Paper Award, CHI 2024 – DynaVis *2024*
Honorable Mention Award, CHI 2024 – ChainForge *2024*
Honorable Mention Award, CHI 2018 – CodeTalk *2018*